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Deepti Singh, Ananda Guha Majumdar, Sunita Gamre, Mahesh Subramanian



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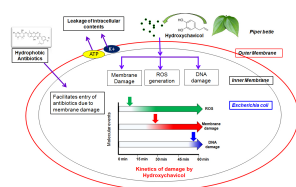
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1 Membrane damage precedes DNA damage in Hydroxychavicol treated *E. coli* cells and
2 facilitates cooperativity with hydrophobic antibiotics.

3 Deepti Singh^{1,2}, Ananda Guha Majumdar^{1,2}, Sunita Gamre¹ and Mahesh Subramanian^{1,2,*}

4 ¹Bio-Organic Division, Bhabha Atomic Research Centre, Trombay, Mumbai. PIN 400 085.

5 ²Homi Bhabha National Institute, Anushaktinagar, Mumbai. PIN 400 094.

6

7 **Correspondence:**

8 Dr. Mahesh Subramanian,

9 Bio-Organic Division, Bhabha Atomic Research Centre, Trombay, Mumbai. PIN 400 085.

10 and

11 Associate Professor, Homi Bhabha National Institute, Anushaktinagar, Mumbai. PIN 400 094.

12 Tel: 022-25590267

13 Email: maheshs@barc.gov.in; barcmahesh@gmail.com

Abstract

Hydroxychavicol (HC), found abundantly in *Piper betle* leaves is credited with antimicrobial property. Previously we had shown HC induces reactive oxygen species mediated DNA damage in bacterial cells. HC also resulted in membrane compromise revealing its pleiotropic effects on cellular targets. The kinetics and exact sequence of events leading to inhibition of growth and cell death in *E. coli* after HC treatment remains poorly understood. We show that sub-lethal concentration (125 µg/mL) of HC causes cellular filamentation within 1 h of treatment, while a higher concentration (750 µg/mL) induces cell breakage. HC treated cells were found to experience oxidative stress as early as 10 minutes, while evidence of membrane damage was apparent at 30 minutes. DNA damage repair genes were found to be activated at 60 minutes. Interestingly, HC-induced cell permeabilization was inhibited and enhanced by external Mg^{2+} and EDTA, respectively, suggesting that HC damages the outer membrane. Kinetic experiments revealed that HC-treated cells underwent oxidative stress, membrane damage and DNA damage in that order. Because gram negative bacteria such as *E. coli* are refractory to several antibiotics due to the presence of the outer membrane, we hypothesized that HC pretreatment would sensitize *E. coli* to hydrophobic antibiotics. Our study reveals for the first time that HC could sensitize bacteria to clinically used antibiotics due to its outer membrane damaging property.

Keywords: Hydroxychavicol; Antibacterial; Drug Resistance; Membrane Damage; Antibiotics

1. Introduction

Rapid emergence of antibacterial resistance has resulted in the possibility of catastrophic failure of current treatment modalities. There is an urgent need to discover new antibacterial compounds coupled with mechanistic understanding of their antibacterial action. This will enable us to better exploit their properties towards eradication of pathogenic bacteria. Natural products have contributed immensely as a drug repository that has helped us combat different health problems, including infectious diseases [1]. *Piper* species have been used in food and medicine for a long time [2]. *Piper betle*, *Piper nigrum* and *Piper auritum* possess a variety of secondary metabolites that have several health benefits [2]. Some of the diseases found to be cured by molecules derived from *Piper species* are skin, liver and stomach ailments, urological problems and inflammation. They have also been proven to be useful as wound healing agent and antipyretic. The phytochemicals and essential oils of *Piper species* have shown to be antibacterial and antifungal [2]. Different parts of these plants like fruits, seeds, leaves, branches, roots and stems possess essential oils and other chemicals of medicinal value [2]. Among the *Piper species*, *Piper betle* is cultivated widely in south and south-east Asia for its heart shaped leaves that are used for as mouth freshener and digestive aid [3]. The leaves contain eugenol, caryophyllene, chavibetol, methyl eugenol and hydroxychavicol (HC) [3, 4]. Among these molecules, HC has been shown to be antibacterial and antifungal [5, 6]. HC was shown to effectively cure fungal infections in two different models employing animals [5]. However, the mode of action of this molecule against microorganisms is not well understood. The molecular target of HC in bacteria is largely unexplored. Plant derived molecules like piperine, epicatechingallate and totarol have also been showed to potentiate the activity of antibiotics [7-9]. Several other plant derived molecules also have been shown to inhibit a variety of multidrug resistant pumps in *S. aureus*

[1]. A detailed evaluation about the mechanism of action of plant derived molecules also resulted in the discovery of clavulanic acid as a β lactamase inhibitor [10], 5'-methoxyhydrnocarpin as a multidrug pump inhibitor [11] and catechins as inhibitors of bacterial toxins [12]. We have previously shown that the antibacterial activity of HC involves reactive oxygen species (ROS) generation, DNA damage leading to cell death [6]. ROS play an important role in the antibacterial action of antibiotics and other antimicrobials [13, 14]. Many antibacterial molecules produce ROS inside bacterial cells and disrupt key processes or damage macromolecules [15, 16]. Antibacterial compounds have also been found to affect processes that result in ROS induction [17]. In either case, ROS has been found to be detrimental to cellular viability. However, phenolic compounds are known to induce cell death by interacting with multiple targets [18]. In case of certain phytochemicals it has been observed that the biological activity is not concentration dependent. They exhibit biphasic dose response or even hormetic response [19]. In this study, we aimed to understand the sequence of events induced by the phenolic compound HC in bacteria leading to cell death. To achieve this we selected ROS generation, membrane damage and DNA damage as the important parameters to be investigated. Time dependent analysis of these parameters in HC-treated *E. coli* cells revealed ROS generation, membrane damage and DNA damage to occur in that order. HC-induced membrane damage occurred through cation abstraction. Based on these findings, we hypothesized that HC-induced membrane damage may facilitate entry of hydrophobic antibiotics inside *E. coli* cells. We found that HC pretreatment sensitized *E. coli* to antibiotics such as rifampicin, vancomycin, novobiocin and kanamycin. Our study highlights the potential of HC as a helper compound that could sensitize bacteria to clinical antibiotics.

2. Materials and methods

2.1. Materials

Luria Bertani broth (LB) and Agar were purchased from Becton Dickinson, USA; Lysozyme, propidium iodide (PI), N-Lauroylsarcosine sodium salt, crystal violet, magnesium chloride, glutaraldehyde, triton X-100, Dichlorofluoresceindiacetate and sodium dodecyl sulfate (SDS) were purchased from Sigma-Aldrich Corporation, USA; EDTA from Merck, Germany; ATP Determination Kit, RNA isolation kit, SYBR green II dye were purchased from Thermo Fisher Scientific, USA; NucleoSpin RNA II from Macherey-Nagel, Germany; Heat and Run DNase from ArcticZymes, Norway; qScript cDNA synthesis kit from Quanta Biosciences, USA; KAPA SYBR Fast qPCR kit from Kapa biosystems, USA; Kanamycin and novobiocin were purchased from Sigma-Aldrich Corporation, USA; Mueller-Hinton (MH) agar, Rifampicin and vancomycin from HiMedia Labs, India; Solvents and other reagents used were of analytical grade or higher and procured locally. The *E. coli* strain used in the study is the K12 wild type strain BW25113.

2.2. Culture growth and drug treatment

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) culture was used for all experiments. HC was dissolved in DMSO and used within 1 week. The final concentration of DMSO in all experiments was 1% (v/v) or less. Vehicle used in all experiments was 1% DMSO (v/v).

2.3. Isolation of HC from *Piper betle* leaves and characterization

HC was isolated from *Piper betle* leaves by solvent extraction and characterized by NMR spectroscopy as described previously [6].

2.4. Monochrome staining with crystal violet

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells of *E. coli* was treated with different concentrations of HC (125, 250, 500, 750 μ g/mL) and vehicle in LB after which representative population from each sample was mounted on clean glass slides, stained with crystal violet (0.5% w/v) for 3 min and observed under a microscope (LSM 780, Carl Zeiss, Germany). For every sample at least 10 fields were observed for every time point. The samples were visually analyzed for the presence of filamented cells.

2.5. Scanning electron microscopy of HC treated cells

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells were treated with different concentrations of HC for 3 h in LB and processed according to the protocol described previously [20]. Cells were observed under a scanning electron microscope (Carl Zeiss, Germany) for surface perturbations induced by HC after coating with gold in an ion coater (KIC-1A, Coxem, Korea).

2.6. Live cell imaging

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells (20 μ L) were mixed with 200 μ l of LB containing 1% agar containing 500 μ g/mL HC. The mixture was mounted on a glass slide coated with 1% agarose. Samples were observed under a microscope (LSM 780 Carl Zeiss, Germany) under confocal mode using a 63 $\times$ objective.

2.7. Comet assay

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells (0.5 mL) 1% glucose-saline solution was treated with HC (500 μ g/mL) for 1 h. After 1 h the cells were washed with saline and pellet obtained by centrifugation was stored at -20 $^{\circ}$ C overnight. The pellets were thawed with 0.5 mL saline solution. The cells from each sample was diluted 1:100 in 0.5% agarose in saline solution and

mounted on a clean glass slide with a base coat of 1% agarose and another layer of 0.5% agarose. A final layer of 0.5% agarose containing 5 µg/mL RNaseA, 1 mg/mL lysozyme, and 0.25% sodium N lauroylsarcosine was overlaid. Once the layer was set at 4 °C for 10 min the slides were incubated at 37 °C for 30 min. The slides were further incubated for 1 h in lysis buffer at RT (2.5 M NaCl, 100 mM EDTA, 10 mM Tris pH 10, 1% sodium lauroylsarcosine, and 1% Triton X-100) followed by a 2 h incubation in enzyme digestion buffer (2.5 M NaCl, 10 mM EDTA, 10 mM Tris pH 7.4, and 0.5–0.6 mg/mL of proteinase K) at 37 °C. Slides were equilibrated for 20 min in electrophoresis tank buffer (300 mM sodium acetate and 100 mM Tris, pH 9). Slides were electrophoresed in the same buffer for 50 min at 12 V. Following electrophoresis, slides were sequentially immersed in 1 M ammonium acetate prepared in ethanol for 20 min, absolute ethanol for 30 min, and 70% ethanol for 10 min. The slides were air-dried at RT. Prior to staining, slides were pretreated with 90 µL of freshly prepared solution containing 5% DMSO in 10 mM NaH₂PO₄. The slides were stained with 40 µL of 1:1000 dilute SYBR II dye for 10 min. The samples slides were scored under a fluorescent microscope (Zeiss LSM 780, Carl Zeiss, Germany) manually using a 63× objective. The data was plotted as the proportion of comets to normal cells in each sample [21]. At least 100 cells were counted for every slide analyzed under the microscope.

2.8. Quantification of gene expression by reverse transcription quantitative real time PCR (RT-qPCR)

RNA was extracted from mid log (OD₆₀₀~0.5 i.e., 10⁷-10⁸ cfu/mL) cells treated with HC (500 µg/mL) for different time periods. The RNA was extracted using NucleoSpin RNA II (Macherey-Nagel, Germany) as per manufacturer's instructions. The RNA was treated with Heat and Run DNase (ArcticZymes, Norway) to remove DNA contamination. Complementary DNA

(cDNA) was synthesized as per the qScript cDNA synthesis kit (Quanta Biosciences, USA) using 1.0 µg of RNA. The qPCR was performed with 10 ng cDNA using the KAPA SYBR Fast qPCR kit (Kapa biosystems, USA) and 0.1 pM of each primer in a 20 µL reaction. The qPCR run and analysis was performed on CFX96 thermal cycler (Bio-Rad, USA). The thermo-cycling conditions included, one cycle of denaturation at 95 °C for 4 min, followed by 45 cycles of denaturing at 95 °C for 2 s and annealing/extension at 60 °C for 40 s. The Ct values were normalized against the *cysG* gene encoding siroheme synthase [22] and the relative gene expression was determined using the $\Delta\Delta C_t$ method [23]. The genes evaluated in the study are listed with their primer details in the Table 1.

2.9. Membrane compromise assay by PI uptake

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells were washed and resuspended in 1% glucose-saline solution and treated with vehicle or HC for 1 h. Cells were washed and resuspended in saline solution containing 30 µM PI. After 15 min incubation, cells were washed with saline solution and fluorescence was quantified in a plate reader with Ex 493 nm and Em 636 nm. The absorbance of each sample at 600 nm was also recorded to ensure the same density of cells across samples. Alternately, a small aliquot of each sample was observed under a fluorescence microscope and images recorded. For evaluation of the kinetics of membrane damage, cells were treated with HC (500 µg/mL). At different time points samples were processed for PI fluorescence as described above.

2.10. Quantification of potassium leakage and ATP leakage

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells (1 mL) were washed, resuspended in saline solution and treated with different concentrations of HC for 1 h. After treatment, the supernate

and cells were separated by centrifugation. The supernate collected (0.9 mL) was diluted in deionized water to 5 mL. The quantity of potassium in supernate was determined by atomic absorption spectrophotometry. In case of ATP determination, the protocol was identical to that of potassium determination. After collection of supernate by centrifugation the intracellular content of the cells were collected in deionized water by sonication (Branson ultrasonic probe sonicator, USA total time: 1 minute comprising 2 s on/5 s off cycles at 25% amplitude) followed by centrifugation ($10,000 \times g$ for 10 min). To quantify the ATP in supernate and cell pellet, 20 μ L of sample was used in a reaction as per the manufacturer's instructions and chemiluminescence was read in a plate reader.

2.11. Rhodamine assay

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells (0.5 mL) were suspended in same volume of 1% glucose-saline solution and treated with HC for 1 h. After 1 h cells were washed, resuspended in 1% glucose-saline solution and stained with 1 μ g/mL Rhodamine 123 for 10 min in the dark [24]. Excess stain was removed by centrifugation and the fluorescence was quantified in a plate reader with Ex 488 nm and Em 530 nm. The absorbance of each sample at 600 nm was also recorded to ensure the same density of cells across samples.

2.12. SDS induced cell lysis assay

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells in LB (1 mL) was treated with HC for 1 h. The cells were washed and resuspended in fresh LB (1 mL). Aliquots (0.2 mL) were pipetted into flat bottom 96 well plates in triplicate. SDS (20 μ L from 1% w/v stock) was added to each well resulting in a final concentration of 0.1%. The absorbance (600 nm) was monitored continuously

for 15 min in a multi-plate reader with reading intervals at 30 s. The absorbance was plotted against time to investigate the rate of lysis of the cells in different samples [25].

2.13. Effect of Mg^{2+} and EDTA on the viability of bacteria treated with HC

The effect of Mg^{2+} and EDTA on HC-induced loss of viability was evaluated by pre-treating the cells with $MgCl_2$ (5 to 20 mM) for 10 min [25]. Briefly 1×10^5 cells were inoculated in 0.2 mL LB with or without $MgCl_2$ in a 96 well plate. After 10 min HC was added. In case of EDTA (0.5 to 10 mM) too pre-treatment was carried out for 10 min following which HC was added. The samples were incubated at 37 °C for 16-18 h. Plates were analyzed at 600 nm. Extent of protection or sensitization on growth was calculated from the absorbance at 600 nm while effect on viability was evaluated by spotting. An aliquot (10 μ L) was spotted on Luria agar plate and incubated for 24 h at 37 °C. The extent of growth was visually observed and image of the plate was acquired.

2.14. Co-detection of ROS and membrane damage in HC treated cells

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells (1.5 mL) were washed and resuspended in 1% glucose-saline solution (1.5 mL). The cells were stained for 5 min with 20 μ M Dichlorofluoresceindiacetate (DCFDA). Cells were washed and distributed in 0.2 mL volumes and treated with HC 500 μ g/mL for different time periods. At the end of incubation the cells were washed, resuspended in 1% glucose-saline solution and stained with 30 μ M PI in the dark for 10 min. Finally the cells were washed again and resuspended in a smaller volume (25 μ L saline solution). A 2 μ L aliquot was mounted on an agarose coated slide and observed under a fluorescence microscope (LSM 780, Carl Zeiss, Germany) for DCFDA and PI fluorescence.

2.15. Sensitivity of HC treated cells to hydrophobic antibiotics

Mid log ($OD_{600} \sim 0.5$ i.e., 10^7 - 10^8 cfu/mL) cells (1 mL) were treated with HC (500, 750 μ g/mL) for 1 h in 1% glucose-saline solution. Post incubation, cells were washed and resuspended in fresh 1% glucose-saline solution. HC treated cells (1×10^6) were inoculated in MH broth (0.2 mL in 96 well plate) in the presence or absence of different concentrations of antibiotics. The plate was incubated at 37 °C for 18 h and analyzed at 600 nm to quantify the extent of growth. Cell viability was evaluated by spotting 10 μ L from each well on a MH agar plate, followed by incubation at 37 °C for 24 h. The extent of growth was visually observed and image of the plate was acquired.

2.16. Evaluation of acute systemic toxicity of HC

The *in vivo* toxicity of HC was evaluated in Swiss mice. Mice were divided into different groups, each group consisting of five mice. Single bolus dose of HC was given by intraperitoneal (i.p.) or oral route in a volume of 0.1 mL in case of i.p and 0.2 mL in case of oral route. HC was prepared in 5% DMSO-saline for i.p. administration and 5% DMSO-edible vegetable oil for oral administration. The different doses of HC administered intraperitoneally were 10, 25, 50, 75, 125, 250 mg/kg body weight and in case of oral administration was 50, 100, 150, 200, 250, 500, 750 mg/kg body weight. Appropriate vehicle treatments without HC were included for comparison. The behavior, food/water intake, weight and survival were monitored for 30 days. The experiments were conducted as per the ethical guidelines and approval of the institute's animal ethics committee.

2.17. Evaluation of acute dermal toxicity of HC

The effect of dermal toxicity of HC was evaluated by primary dermal irritation test by application of a formulation containing HC and observation of animals for 72 h for irritation.

Wistar rats of either sex were shaved of the fur in the trunk region and the skin was abraded (1 inch $\times$ 1 inch approximately). The formulation containing HC was applied on a patch which was placed on skin and taped. The formulation used was petroleum jelly/liquid paraffin (2:1), containing 1% and 2.5% HC (w/w). Further it was covered with an impervious tape. After 24 h the patch was removed and the rats were observed for erythema and edema. The rats were observed again after 48 h of removal of the patch. The experiments were conducted as per the ethical guidelines and approval of the institute's animal ethics committee.

2.18. Data analysis

Experiments were conducted in triplicates and repeated thrice. The data presented are mean $\pm$ Standard deviation (SD) of the three independent experiments. Significant differences between the treatments were evaluated using SD and one-way analysis of the variance (ANOVA) using SPSS 2.0. The data between the two specific different treatments were compared statistically using Student's t-test. The differences were considered significant at $P < 0.05$.

3. Results

3.1. HC treatment results in filamentation of *E. coli* cells

We had previously determined the MIC and MBC of HC (Structure provided in figure 1) against *E. coli* to be 250 $\mu\text{g/mL}$ and 500 $\mu\text{g/mL}$, respectively [6]. In the current investigation an understanding of the molecular basis of the different phenotypes induced by HC was envisaged. Upon treatment of *E. coli* cells with different concentrations of HC, changes in the cellular morphology were observed. As shown in figure 2A, filamentation was observed in HC-treated samples. The filamentation induced by 125 $\mu\text{g/mL}$ was overcome by the cells within 2 h (figure 2A), while at 250 $\mu\text{g/mL}$ the filamentation persisted beyond 3 h (figure 2A). Incidentally, at the

highest concentrated tested (750 $\mu\text{g/mL}$) filamented cells were few. Similar result was obtained when HC treated samples were observed by scanning electron microscopy (SEM). Filamentation was observed upon HC treatment at 250 $\mu\text{g/mL}$ and 500 $\mu\text{g/mL}$ by SEM (figure 2B). At 750 $\mu\text{g/mL}$, many broken cells were observed corroborating the absence of filamented cells. Quantification of the mean cell lengths and proportion of different phenotypes after treatment with HC is presented in figure 3A-B. Live cell imaging of HC treated *E. coli* cells (up to 3 h) revealed significant inhibition of cell growth as well as division at 500 $\mu\text{g/mL}$ (figure 4). At 750 $\mu\text{g/mL}$, the cells shrunk and lost granularity indicating broken cells and could not be observed properly under the microscope during the live cell imaging experiment. This could also be seen from the SEM picture of 750 $\mu\text{g/mL}$ treated cells (figure 2B). This indicated the death of cells caused by 750 $\mu\text{g/mL}$ is sudden and the target is likely to be the membrane of the cell as evidenced from SEM and live cell imaging data.

3.2. HC treatment induces DNA damage and DNA damage repair genes

Physical damage to DNA in HC treated *E. coli* cells was visualized employing comet assay. No DNA damage was observed in cells treated with vehicle or 125 $\mu\text{g/mL}$, while significant DNA damage was observed at 250, 500 and 750 $\mu\text{g/mL}$ (figure 5A). Evaluation of the kinetics of the gene expression of DNA damage repair genes employing real time PCR revealed significant up regulation of *nth*, *nei*, *recN* and *dinB* genes. While *nth* was found to be upregulated after 40 min (figure 5B), *nei* and *recN* were found to be up regulated at 60 min. The up regulation of *dinB* was found to be at 120 min. The maximum up regulation of *nth* was found to be at 60 min. RecN is a member of the Structural Maintenance of Chromosome (SMC) class of proteins [26]. Oxidative DNA damage is repaired by Nth and Nei [27], while DinB is a DNA polymerase involved in error prone DNA repair [28]. These data indicate oxidative DNA damage repair response to be

optimally activated at 60 min. Interestingly, RecN has been recently implicated in filamentation in response to DNA damage [29], suggesting that induction of recN in response to HC treatment might have a bearing on HC-induced cellular filamentation.

3.3. HC induces membrane damage

We investigated the induction of membrane damage employing the PI uptake assay. We treated *E. coli* cells with HC and evaluated membrane damage by PI uptake. HC treated cells exhibited PI uptake after 1 h of treatment (figure 6A-B). Evaluation of PI uptake at MBC revealed significant uptake at 15 min and thereafter (figure 6C-D). This prompted us to check potassium and ATP leakage as markers of membrane damage. HC induced significant potassium leakage from cells (figure 7A). Kinetic analysis revealed potassium leakage begins at 30 min (figure 7B) after HC treatment. The lowest concentration of HC that induced leakage of ATP from *E. coli* cells was found to be 500 $\mu\text{g/mL}$ after 1 h of treatment (figure 8A). Similar to potassium leakage, cells treated with HC revealed significant ATP leakage at 30 min and beyond (figure 8B).

3.4. HC induces membrane hyperpolarization

A key feature of an intact bacterial membrane is the membrane polarity, with the intracellular milieu being relatively negative. Membrane damage results in ionic equilibration, causing membrane depolarization [30]. However hyperpolarization has also been reported to accompany loss in viability [31]. Since membrane damage was detected after HC treatment we evaluated the membrane potential employing Rhodamine 123 [32]. As shown in the figure 9A, significant membrane hyperpolarization was observed after 1 h of HC treatment. A time kinetic analysis revealed membrane hyperpolarization induced by HC (500 $\mu\text{g/mL}$) evident at 15 min after which it increased till the maximum time point observed (1 h) (figure 9B).

3.5. Susceptibility of HC treated cells to membrane lysing agents

When grown in the presence of a mild membrane lysing agent (0.1% SDS), higher rate of cell lysis was observed in case of HC (500 and 750 µg/mL) treated cells compared to vehicle treated cells. This showed that HC treated cells exhibited weakened membrane and were susceptible to treatment that would not affect healthy cells (figure 10).

3.6. Effect of Magnesium and EDTA on HC-induced toxicity

There are different mechanisms by which membrane is damaged by antibacterial agents [33]. One of the mechanisms by which a small organic molecule might compromise a bacterial membrane is by stripping Mg^{2+} leading to collapse of the same [34]. We evaluated the effect of Mg^{2+} on HC-induced growth inhibition and cytotoxicity in *E. coli*. External supplementation of Mg^{2+} protected cells from HC, resulting in higher growth (figure 11A). Conversely, external addition of EDTA resulted in lower growth in presence of HC (figure 11B). External Mg^{2+} had an effect on MBC as revealed from the spot test (figure 11C). However, EDTA only affected the extent of growth (lesser growth compared to HC alone) and not the MBC (figure 11D). For e.g., no sensitization was observed in case of HC (200 µg/mL and 250 µg/mL) treatment in EDTA supplemented population (figure 11D). We further observed that the effect of Mg^{2+} on HC-induced inhibition of growth did not hold true for gram positive bacterium *S. aureus*. External supplementation of Mg^{2+} did not have any effect on extent of growth as well as cytotoxicity induced by HC in *S. aureus* (Supplementary figure 1). This data, coupled with the effects of Mg^{2+} and EDTA on HC-induced membrane permeabilization suggests that HC destabilizes the outer membrane through cation abstraction.

3.7. Time kinetics of damage induced by HC

Because multiple targets were damaged after HC treatment, it is imperative to understand the relative temporal order of the events. We devised a method to simultaneously detect ROS generation and membrane damage after HC treatment, using DCFDA and PI uptake as markers of ROS generation and membrane damage, respectively. DCFDA positive cells were detected as early as 10 min post HC treatment, while significant PI uptake was observed earliest at 30 min (figure 12A). This data also agrees with the potassium and ATP leakage data. Quantitative analysis of the proportion of DCFDA positive, PI positive and double positive cells (figure 12B) revealed DCFDA positive cells appeared at 10 min and peaked at 15 min, with significant proportion of cells remaining DCFDA positive even at 60 min. Significant proportion of PI positive was detected at 30 min and 60 min.

3.8. Pretreatment with HC sensitizes *E. coli* towards hydrophobic antibiotics

To test if HC-induced membrane damage would enhance the sensitivity of *E. coli* to hydrophobic antibiotics, cells were pretreated with HC (250, 500 and 750 µg/mL) for 1 h. The pretreatment alone did not affect the growth and viability. But when challenged with different antibiotics, the pretreated cells exhibited higher sensitivity compared to cells that were not pretreated with HC. Compared to cells that were pretreated with vehicle HC pretreated cells exhibited less growth as well as viability with each of the antibiotics tested (figure13A-H).

3.9. Acute toxicity of HC on rodents

The toxicity of HC was evaluated employing Swiss mice. HC was administered either orally or intraperitoneally. HC was observed to be safe up to 75 mg/kg body weight when administered intraperitoneally and 750 mg/kg body weight when administered orally. At concentrations higher than 75 mg/kg body weight, when administered intraperitoneally the mice died within 24 h. At

non lethal doses, the mice did not show any abnormal behavior in water or food intake, grimace or gait and all mice survived beyond 30 days. In the case of acute dermal toxicity in rats, at both the concentrations employed (1% and 2.5% w/w HC) no significant erythema or edema was observed compared to untreated group or the group treated with formulation without HC.

4. Discussion

The generation of ROS is considered a key effector of cell death in mammalian as well as prokaryotic cells. Many plant derived molecules have been shown to function against cancer cells by tipping the balance of oxidative stress in the cell [35]. Prokaryotic cells, including infectious bacteria, are susceptible to oxidative stress and molecular damage due to them. ROS has also been shown to induce apoptosis-like cell death in bacteria [15, 36]. A common ROS mediated pathway underlying bacterial cell death treated with various antibiotics have been repeatedly proposed and contested [16, 37, 38]. This shows that the mechanism and importance of ROS generation by antibacterial agents is not very well understood. Infectious bacteria have steadily acquired resistance to different clinically used antibiotics by a variety of mechanisms, resulting in frequent failure of established treatment regimens. As an endeavor towards discovery of new antibacterial molecules, we had previously reported the antibacterial effect of HC and its mechanism of action [6].

In the context of ROS, though oxidative stress was detected within 10 minutes of exposure to HC, induction of DNA damage repair genes, *nth* and *nei* that are typically involved in the repair of oxidative damage peaked at 60 minutes. Plant derived molecules have been known to exhibit cytotoxicity through complex multiple pathways affecting different targets [18]. Hence we investigated if membrane compromise could be a phenomenon contributing to cell death by HC in addition to ROS induced DNA damage. Outer membrane of gram negative bacteria physically

excludes many antibacterial compounds. Poor efficacy of plant derived antibacterial compounds in gram negative bacteria compared to gram positive bacteria has been recorded previously [39]. In this context a phytochemical that damages the membrane is desirable. The ability of HC to damage membrane revealed by uptake of PI and leakage of intracellular ATP and potassium was thus interesting. Because no oxidative lipid end products were observed [6] we investigated the mechanism underlying this membrane compromise induced by HC in *E. coli*. Our results essentially demonstrated that HC destabilizes the outer membrane through cation abstraction. Membrane active phytochemicals have been reported previously and their mechanism of action has been found to be diverse [40]. A general mechanism common for different phenolic phytochemicals affecting the membrane bilayer of bacteria thereby compromising functions of different proteins was proposed recently [41]. A similar phenomenon occurring in case of HC is most likely. However the high amount of oxidative stress detected after HC treatment is also intriguing. Hence, for all the different findings those support a common mechanism for phenolic antibacterials, subtle differences do exist in the events that lead to death of a bacterial cell. It is important to understand these differences from the point of scientific curiosity as well as to better exploit these molecules as antibacterial agents.

Outer membrane permeabilization has been identified as a strategy to make bacteria susceptible to different antibiotics [42, 43]. We provide strong evidence in our investigation towards membrane compromising activity of HC (in the form of susceptibility to SDS, sensitization by EDTA, and protection by Mg^{2+}) resulting in hyperpolarization of the membrane. Further this event alone may not explain the antibacterial activity as oxidative stress in the cells and DNA damage were also detected. Their contribution towards cell death could not be undermined. The cell death thus could be a result of damage to multiple targets after HC treatment. To our

knowledge, events at the membrane level have not been given sufficient priority as a potential target to kill bacteria. Recently, a lot of focus has been on the events occurring at the membrane after treatment with different antimicrobial compounds [33, 44, 45]. To exploit the membrane compromise induced by HC we tested if pretreatment with HC could sensitize the cells to clinically used hydrophobic antibiotics. Cells pretreated with HC exhibited higher sensitivity to clinically used antibiotics such as vancomycin, novobiocin, rifampicin and kanamycin. Kanamycin, an important aminoglycoside though strictly not a hydrophobic antibiotic, does not permeate the outer membrane freely and requires presence of OmpC and OmpF proteins that form pores in outer membrane of gram negative bacteria for its entry into the cells [46]. Hence we included this antibiotic in our study. Previously, phytochemicals such as thymol and gallic acid have also been observed to permeabilize membrane of gram negative bacteria sensitizing them to clinically used antibiotics. Thymol and gallic acid are antibacterial at concentrations greater than 600 and 700 $\mu\text{g/mL}$ respectively against *E. coli* [43]. Upon co-inhibition of *E. coli* cells with thymol with piperacillin, a ~2 fold increase in zone of inhibition was observed compared to piperacillin alone. In case of gallic acid ~1.4 fold increase in zone of inhibition was observed. When tested with another antibiotic cefoperazone the fold increase in sensitivity was less than 1.5 fold for both thymol and gallic acid [43]. However co-incubation in liquid medium resulted in 4 fold reduction in MIC in case of both thymol and gallic acid with cefoperazone against *E. coli* [43]. Though different phytochemicals have been reported previously to sensitize bacterial cells to clinically used antibiotics, the mechanisms vary vastly. The general mechanisms by which cooperativity is achieved are 1. Multi target effects by which cooperating partners affect different targets in the cell. Totalrol, a diterpene that profoundly affects respiration potentiated the effect of methicillin on *S. aureus* by affecting PBP2a expression [9].

2. Pharmacokinetic or physiochemical effect i.e. improvement of solubility or bioavailability. Solubility and oral bioavailability of hypericin is enhanced by mixtures containing procyanidin B2 and hyperocidin. Pharmacokinetic synergy also results from constituents that inhibit enzymes that convert drugs into excretable or inactive forms, or that activate enzymes that convert pro-drugs into active forms [47]. 3. Targeting a specific resistance mechanism e.g. efflux pump inhibition. Bergamottin and coumarin epoxides found in grape fruit oil enhanced the sensitivity of MRSA to norfloxacin by greater than 10 fold due to efflux pump inhibition [48]. Piperine analogs also inhibit efflux pumps contributing to their synergy with antibiotics [7].

To this end our work extensively establishes the mechanism of action of the antibacterial activity of HC and kinetics of damage to different targets in the cell. The upshot of the mechanistic understanding is the use of HC as a helper compound with antibiotics that do not work against gram negative bacteria. This sensitization has a very important potential towards combating different gram negative bacteria such as *Klebsiella spp*, *Pseudomonas spp*, *Proteus spp* etc. Further this finding also augments the application of hydrophobic antibiotics towards combating infections. Considering HC functions at a higher concentration it may be an appropriate candidate to be used as a topical antibacterial agent. Coupled with the wound healing activity of HC [49] and the non toxic nature, it could potentially be used to treat wounds in diabetic patients after further investigation.

Conflict of Interest

None to declare

Funding

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Author contributions

DS and AGM were involved in collection of data. SG isolated HC used in the study. MS conceptualized the work, analyzed the data and wrote the manuscript. All the authors have approved the submitted manuscript.

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Figure legends

Figure 1. Structure of Hydroxychavicol.

Figure 2. Phenotypic changes induced by HC on *E. coli*. (A) Mid log cells were treated with different concentrations of HC for the indicated time. Samples were smeared on glass slides, stained with crystal violet and observed under the microscope. (B) Mid log cells were treated with HC for 3 h and processed to be observed under a scanning electron microscope.

Figure 3. Quantification of phenotype induced by HC treatment. (A) Mid log cells were treated with HC for 3 h and the phenotype captured by scanning electron microscopy. The cell lengths were measured for each treatment with the associated software and plotted. (B) The proportion

of the cells bearing different phenotypes after treatment with HC for 3 h. The values given are mean $\pm$ SD.

Figure 4. Effect of HC on the growth of *E. coli* cells by live cell imaging. Mid log cells were mixed with HC in molten Luria agar and overlaid on agarose coated slide and observed continuously up to 4 h for the effect on growth. (a-c) Vehicle treated samples: 0, 1 and 2 h, respectively. (d-h) HC (500 μ g/mL) treated samples: 0, 1, 2, 3 and 4 h, respectively.

Figure 5. DNA damage induced by HC. (A) Mid log cells were treated with HC and processed for comet assay. The number of comets in each sample was analyzed under the microscope and expressed as a proportion to that of normal cells. (B) Mid log cells were treated with HC (500 μ g/mL) for different time points. RNA was extracted and cDNA prepared, followed by qPCR. Relative gene expression was obtained and plotted. The values given are mean $\pm$ SD. *indicates significant difference at $p < 0.05$ (unpaired Student's t test). The experiment was done in triplicate and repeated thrice.

Figure 6. Membrane damage induced by HC in *E. coli*. Mid log phase cells were treated with HC for 1 h after which the PI uptake was visualized by (A) microscopy and (B) quantified in a fluorescence plate reader. In a time kinetic assay, cells treated with HC (500 μ g/mL) were processed at different time points for PI uptake either for visualization by microscopy (C) or quantified in a fluorescence plate reader (D). The values given are mean $\pm$ SD. *indicates significant difference at $p < 0.05$ (unpaired Student's t test). The experiment was done in triplicate and repeated thrice.

Figure 7. Estimation of potassium leakage from HC treated *E. coli*. Mid log cells were treated with (A) HC for 1 h (B) HC (500 μ g/mL) for different time periods after which the amount of

potassium leaked out of the cells was quantified by atomic absorption spectrometry. The values given are mean $\pm$ SD. *indicates significant difference at $p<0.05$ (unpaired Student's t test). The experiment was done in triplicate and repeated thrice.

Figure 8. Estimation of ATP leakage from HC treated *E. coli*. Mid log cells were treated with (A) HC for 1 h (B) HC (500 μ g/mL) for different time periods after which the amount of ATP leaked out of the cells in supernate and residual ATP in the pellets was quantified by chemiluminescence. The values given are mean $\pm$ SD. *indicates significant difference at $p<0.05$ (unpaired Student's t test). The experiment was done in triplicate and repeated thrice.

Figure 9. Effect of HC treatment on the state of energy metabolism. Mid log cells were treated with (A) HC for 1 h (B) HC (500 μ g/mL) for different time periods after which the fluorescence of rhodamine 123 was quantified in a plate reader. The values given are mean $\pm$ SD. *indicates significant difference at $p<0.05$ (unpaired Student's t test). The experiment was done in triplicate and repeated thrice.

Figure 10. HC treated cells exhibit higher rate of lysis. HC treated *E. coli* were washed and re-suspended in LB. The rate of lysis was monitored continuously in the presence of 0.1% SDS at 600 nm in a plate reader. The experiment was conducted in triplicate and repeated thrice. As a representation the mean values are provided.

Figure 11. Effect of magnesium and EDTA on HC mediated growth inhibition of *E. coli*. Mid log cells were pretreated with Mg^{2+} and EDTA before addition of HC. Extent of growth and viability was evaluated by absorbance and plating, respectively. Heat map based on the extent of growth in presence of Mg^{2+} (A) and EDTA (B) was generated. (C-D) Effect of Mg^{2+} and EDTA, respectively on the viability of *E. coli* treated with HC.

Figure 12. Simultaneous detection of ROS and membrane damage in HC treated cells. ROS by DCFDA and membrane damage by PI was detected simultaneously by pre labeling cells with DCFDA before HC treatment. After HC treatment the cells were labeled with PI and observed under the fluorescence microscope (A). The scale bar in the figure represents 10 μ m. Different fields were scanned and the unstained, DCFDA positive, PI positive and double positive cells were counted. For each sample at least 500 cells were counted before plotting the graph (B). The values are mean $\pm$ SD. The experiment was conducted in triplicate and repeated thrice.

Figure 13. Sensitization of *E. coli* by HC to hydrophobic antibiotics. Cells were pretreated with HC for 1 h and washed before exposure to different antibiotics. Extent of growth and viability was evaluated by absorbance and plating, respectively. Heat map (A, C, E, G) based on the extent of growth was generated. Effect of the treatment on viability was evaluated by spotting (B, D, F, H) an aliquot after 18 h incubation on MH agar. The growth on plates was observed after 24 h of spotting.

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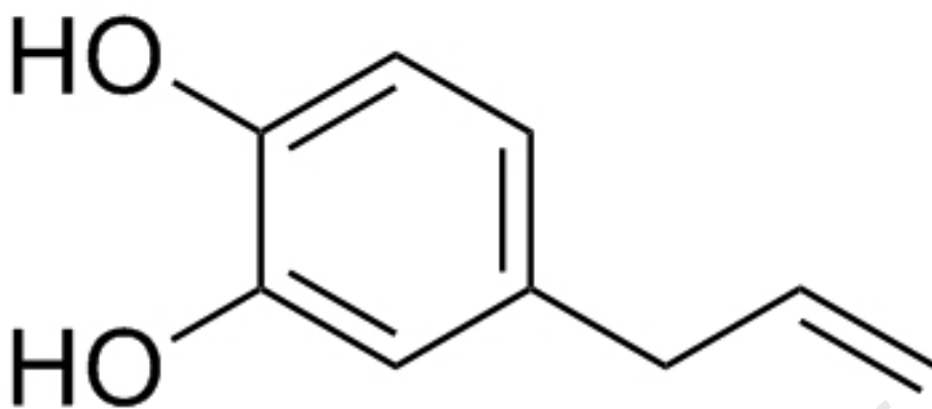
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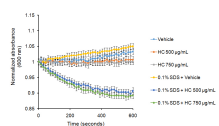
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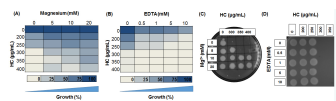
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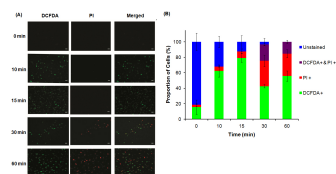
List of Primers used to analyze expression of the DNA damage response genes after treatment with Hydroxychavicol.

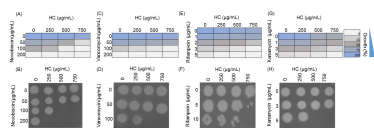
Sr. No.	Gene Name	NCBI Gene ID	5' Primer	3' Primer	Amplicon Size (bp)
1	uvrA	948559	CGGGACGTTTCAG CTTTA	GCCTCTTCGATGGT CATATC	205
2	sulA	947335	TGGGCTACCCTTA ACGAA	GCTTACCGGACGC ATAATAA	205
3	recN	947105	CGATCCCAACCGA CTATTTG	GTGCCTGCTGATG ATGTT	194
4	nth	947122	TTGAACTGGGCGT TGAAG	CGTGTCGACAGCA ATAGTC	224
5	nei	945320	CCAACGACTTAAC GCTCTAC	ATGCGTGGTCAGT TGTTT	182
6	dinB	944922	CGCCTCCGACATG AATAA	GCATCACCAGATC ACACTT	179
7	cysG	947880	GCAGATGCAGGCA TGTTA	CATCGACCACGTT ACAGAAG	157

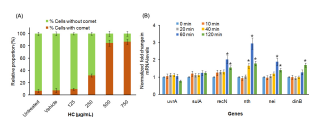


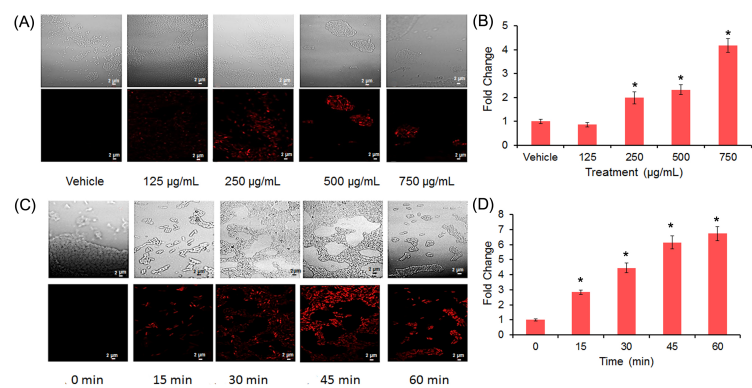


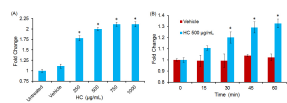


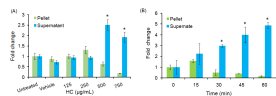


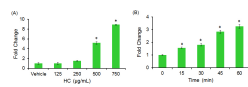


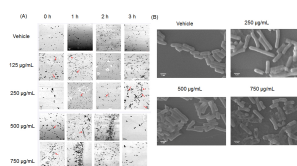


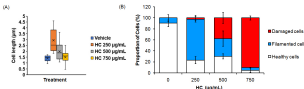


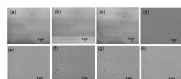












Highlights

Hydroxychavicol causes membrane damage in *E. coli* by stripping magnesium ions.

The kinetics of the damage to various targets was elucidated.

Hydroxychavicol induces oxidative stress followed by membrane and DNA damage.

Hydroxychavicol sensitizes *E. coli* to clinically used antibiotics.

Conflict of Interest

None

Journal Pre-proof